

the role of packaging from a passive participant to an active enabler. It was then said that packaging solutions could provide contributions to the final product by providing introduction of a “*half technology node generation*” ahead of schedule.

It soon became clear that this first step in novel packaging technology was only the beginning of an intense integration between dice and packages in providing the most viable and timely system-oriented solutions.

In 2014 the various teams of the International Technology Roadmap for Semiconductors (ITRS) effort—realizing that many transformations in IC manufacturing were underway—warned that major changes were coming in the electronics industry, such as new IC/package roles. The ITRS identified that the relationship between systems, integrated circuits, and packages was becoming symbiotically intermingled going forward.

Due to this realization the ITRS reassembled the seventeen highly specialized technology working groups (TWGs) existing at the time into seven integrated focus teams (IFTs). The first reports of this new organization were published in the 2015 ITRS2.0. Among the newly created chapters a forward-looking chapter entitled “Heterogeneous Integration” was published and for the first time provided a novel view of the many integrated packaging and IC solutions coming in the next 15 years. It was expected that initially *homogeneous integration* of electronics components (i.e., chips based on similar silicon technologies) were going to populate the system in package world and then evolve into more complex ones in the subsequent 10-15 years. Within this projected timeline an era of truly heterogeneous integration of ICs (created with fundamentally different technologies) was to occur starting with the year 2025; as such, this process is just beginning.

As predicted by the ITRS in 2015 and subsequently by the IRDS, the interaction and symbiosis between ICs and packages has continued to synchronically and homogeneously increase and as a result multi-chip modules have now become fundamental building blocks in the manufacture of any high-performance electronics system. Many companies have created their own approach and *self-defined nomenclature*, and, even though some differences exist, it can be shown that fundamentally these are all variations on the same technology.

The Approach to the Executive Packaging Tutorial

This EPT has been assembled in three parts. Each one of them is essentially “self-contained.” This approach allows the knowledgeable reader to advance to the section of their interest without necessarily reading several of the previous sections. This approach comes at a price since it creates some intrinsic redundancy but provides the benefit to the reader of “self-contained” sections. Therefore, the readers should be aware that this is not an editorial failure but it is motivated by a necessary level of redundancy.

While the main purpose of Part 1 is to outline the features and evolution of packaging technologies leading to system in package for HPC and AI, Part 2 continues the historical progression with respect to system-on chip-technologies, system in package (SiP), and system in board (SIB) and their evolution, integration, and applications’ trends for future HPC and data center requirements. A more extensive analysis of SoC is provided in the 2024 IRDS Executive Summary, and it is recommended that the two documents are evaluated side by side.

In order to make the content more easily readable additional details are provided in the appended material for further insights. Readers can refer to each of the EPT’s appended material for in depth information and discussion.

Key Trends

Semiconductor packaging is evolving rapidly due to the increasing demands for smaller, faster, and more energy-efficient electronics products. Associated with these demands are significant trends that are expected to shape the future of semiconductor packaging landscape for consumer products, drive innovation and enable the next generation of electronic devices at large. See Table 1.

Table 1. Key Trends and Advanced Packaging Evolution

Trend	Impact
Homogeneous Integration	2.5D and 3D Packaging: These technologies enable stacking multiple chips on top of each other or side by side, leading to better performance and lower power consumption. 2.5D involves placing chips side by side on an interposer, while 3D involves stacking them vertically. - Chiplet Architecture: Instead of designing a single large monolithic chip, chiplets allow smaller, function-specific chips to be combined, offering flexibility and cost advantages. - Fan-Out Wafer-Level Packaging (FOWLP): This packaging technique offers a higher level of miniaturization and better thermal performance, making it suitable for high-performance computing and mobile devices.
Heterogeneous Integration	Integration of Diverse Technologies: Combining different types of chips (e.g., analog, digital, non-silicon based and photonic) into a single package allows for more complex and capable systems. This trend is crucial for applications like artificial intelligence (AI), the internet of things (IoT), and 5G.
System-in-Package (SiP)	Miniaturization of Entire Systems: SiP packages multiple components (e.g., processors, memory, sensors) into a single module, enabling compact devices with enhanced functionality. This trend is particularly important for wearable technology and IoT devices.
Thermal Management Innovations	Advanced Cooling Solutions: As chips become more powerful, managing heat dissipation is critical. Techniques like integrated liquid cooling, improved thermal interface materials, and innovative packaging designs are gaining importance.
Materials Advancements	- Use of New Materials: The shift to new materials like silicon carbide (SiC) and gallium nitride (GaN) is enabling higher efficiency and better performance, particularly in power electronics and RF applications. - Bio-compatible Materials: For medical devices and wearables, there's a growing demand for packaging materials that are safe for long-term contact with the human body.
Reliability and Testing	- Improved Testing Techniques: As packaging becomes more complex, so does the need for advanced testing methods to ensure reliability, especially for critical applications in automotive, aerospace, and healthcare. - Enhanced Reliability: Packaging designs are increasingly focusing on improving reliability under harsh conditions, such as high temperatures and mechanical stress, which is critical for automotive and industrial applications.
Sustainability and Cost Efficiency	- Eco-friendly Packaging: The semiconductor industry is pushing towards more sustainable practices, including the use of recyclable materials, reduction of hazardous substances, and energy-efficient manufacturing processes. - Cost-Effective Manufacturing: The ongoing trend towards cost reduction will drive innovations in packaging processes that can reduce overall manufacturing costs without compromising performance.
AI and Machine Learning in Packaging Design	Optimization through AI: Artificial intelligence and machine learning are increasingly being used to optimize packaging design and manufacturing processes, leading to better performance and reduced time-to-market.
Photonics Integration	Photonic Packaging: With the growing importance of optical communication, integrating photonics with traditional electronic chips in a package is becoming more common, enabling faster data transfer and improved signal integrity.
Security Features	Hardware-Level Security: As cyber threats increase, embedding security features directly into semiconductor packages is becoming a trend, especially for applications in defense and secure communications.

The Special Role of Packaging in High Performance Computing

High-performance computing (HPC) is at the forefront of advancing scientific research, artificial intelligence, and complex simulations. For this reason, the role of semiconductor packaging in HPC is critical, as it directly impacts the performance, power efficiency, and scalability of HPC systems. Table 2 presents an overview of how packaging contributes to HPC.

Table 2. Packaging Contributions to High-performance Computing

Packaging Contribution to HPC	Description
Performance Enhancement	- Interconnect Density: Advanced packaging technologies like 2.5D and 3D integration enable higher interconnect density, reducing the distance between components. This minimizes latency and improves data transfer speeds, which are crucial for HPC workloads. - Reduced Parasitics: By using advanced packaging techniques that place components closer together, the parasitic effects such as resistance, capacitance, and inductance are reduced. This enhances signal integrity and overall system performance.
Thermal Management	- Efficient Heat Dissipation: HPC systems generate significant heat due to the high power consumption of CPUs, GPUs, and other processing units. Advanced packaging solutions integrate efficient thermal management techniques, such as thermal interface materials (TIMs), heat spreaders, and liquid cooling, to maintain optimal operating temperatures. 3D Packaging Challenges: While 3D stacking increases performance by reducing interconnect lengths, it also poses challenges in heat dissipation. Packaging plays a crucial role in addressing these thermal challenges through innovative cooling solutions.
Power Efficiency	- Power Delivery Networks (PDNs): Packaging technology directly influences the efficiency of power delivery to the components. High-performance packaging solutions ensure low-resistance and low-inductance PDNs, which are vital for stable and efficient power delivery in HPC systems. - Energy Efficiency: Advanced packaging technologies like wafer-level packaging (WLP) and flip-chip can optimize energy efficiency by minimizing power loss and improving signal integrity.
Scalability and Integration	- Heterogeneous Integration: HPC systems often require the integration of diverse components, such as CPUs, GPUs, memory, and specialized accelerators. Packaging plays a crucial role in heterogeneous integration, enabling the combination of different technologies (e.g., silicon, GaN, InP) within a single package. - Chiplets: The use of chiplet-based architectures, where multiple smaller dies are packaged together, allows for scalable and modular HPC systems. This approach facilitates customization and rapid iteration of HPC designs.
Data Bandwidth and Communication	- High-Bandwidth Memory (HBM): Advanced packaging is essential for integrating HBM with processors in HPC systems. The close proximity of memory and processors, enabled by 2.5D and 3D packaging, allows for extremely high data bandwidth, which is crucial for HPC workloads. - Optical Interconnects: As data bandwidth requirements continue to grow, packaging technology is evolving to support optical interconnects. These interconnects offer higher data rates and lower latency compared to traditional electrical interconnects, which is vital for HPC performance.
Reliability and Durability	- Robust Packaging Materials: HPC systems operate under extreme conditions, including high temperatures and constant high loads. Packaging materials must be durable and reliable to ensure long-term operation without failure. - Failure Mitigation: Advanced packaging techniques include features that

Packaging Contribution to HPC	Description
	mitigate the risks of failure, such as redundant interconnects, advanced sealing methods, and protection against electromagnetic interference (EMI).
Cost and Manufacturing Considerations	- Economies of Scale: As HPC systems scale in complexity and production volume, packaging solutions must also scale efficiently. Advanced packaging technologies can help reduce costs through innovations in manufacturing processes, such as wafer-level processing and automated assembly. - Yield Management: In HPC, where the performance of individual components is paramount, packaging plays a role in maximizing yield by reducing defects and ensuring that only high-performance components are used.

In summary, packaging technology is a critical enabler for HPC, influencing everything from performance and power efficiency to scalability and reliability. As the demands of HPC continue to grow, so too will the importance of advanced semiconductor packaging. Impacts on future directions include emerging packaging technologies, with innovations such as fan-out wafer-level packaging (FOWLP) and embedded multi-die interconnect bridge (EMIB) are being explored to further enhance the performance and integration capabilities of HPC systems. Quantum Computing will be impacted too, as quantum computing becomes more viable, packaging will play a crucial role in integrating quantum processors with classical computing components, ensuring coherence and minimizing interference.

Silicon as a Packaging Substrate

Silicon is increasingly being used as a packaging substrate material in advanced semiconductor packaging technologies. Traditionally, packaging substrates were made from materials like organic laminates (e.g., FR4), ceramic, or glass. However, the use of silicon as a packaging substrate offers several unique advantages that make it well-suited for high-performance and high-density applications. Below is an overview of the role of silicon as a packaging substrate and Table 3 shows several advantages of using silicon as a packaging substrate.

Table 3. Advantages of Silicon as a Packaging Substrate

Silicon Packaging Substrate Attributes	Advantages
High Thermal Conductivity	- Efficient Heat Dissipation—Silicon has a high thermal conductivity (around 150 W/m·K), making it effective at dissipating heat generated by high-performance chips. This is crucial for maintaining the reliability and longevity of semiconductor devices, particularly in applications like high-performance computing (HPC) and data centers. See Figure 1 for an illustration of thermal conductivity. - Thermal Management Integration—Silicon substrates can be integrated with microfluidic channels or other advanced cooling solutions, further enhancing thermal management.
High-Density Interconnects	- Fine-Line Routing—Silicon allows for the creation of fine-line routing and high-density interconnects, which are essential for advanced packaging technologies like 2.5D and 3D integration. This capability supports the increasing need for higher pin counts and finer pitches in modern ICs. - Through-Silicon Vias (TSVs)—Silicon substrates can be fabricated with TSVs, enabling vertical electrical connections in 3D stacked architectures. TSVs reduce the length of interconnects, thereby minimizing signal propagation delay and improving overall system performance.
Mechanical Properties	Dimensional Stability—Silicon has excellent dimensional stability and a low coefficient of thermal expansion (CTE), which is closely matched to that of silicon chips. This reduces the risk of thermal

Silicon Packaging Substrate Attributes	Advantages
	mismatch and mechanical stress between the substrate and the chips, leading to improved reliability. • High Modulus—Silicon’s high modulus (stiffness) provides structural integrity to the package, reducing warping and deformation during the assembly and operation processes.
Electrical Performance	• Low Electrical Losses—Silicon substrates exhibit low electrical losses, which is beneficial for high-frequency applications. This characteristic supports the efficient transmission of high-speed signals, which is important in communication, RF, and high-speed computing systems. • Integrated Passive Devices (IPDs) —Silicon substrates can be used to integrate passive components like capacitors, resistors, and inductors directly onto the substrate, enhancing the overall performance and reducing the footprint of the package.
Compatibility with Silicon-based Devices	• Monolithic Integration—Silicon substrates allow for the potential monolithic integration of silicon-based devices and circuits directly onto the packaging substrate. This integration simplifies the overall design and can lead to more compact and efficient systems. • Ease of Manufacturing—Silicon is already widely used in semiconductor manufacturing, which means that the infrastructure and processes for handling and processing silicon are well-established. This compatibility facilitates the adoption of silicon substrates in advanced packaging.

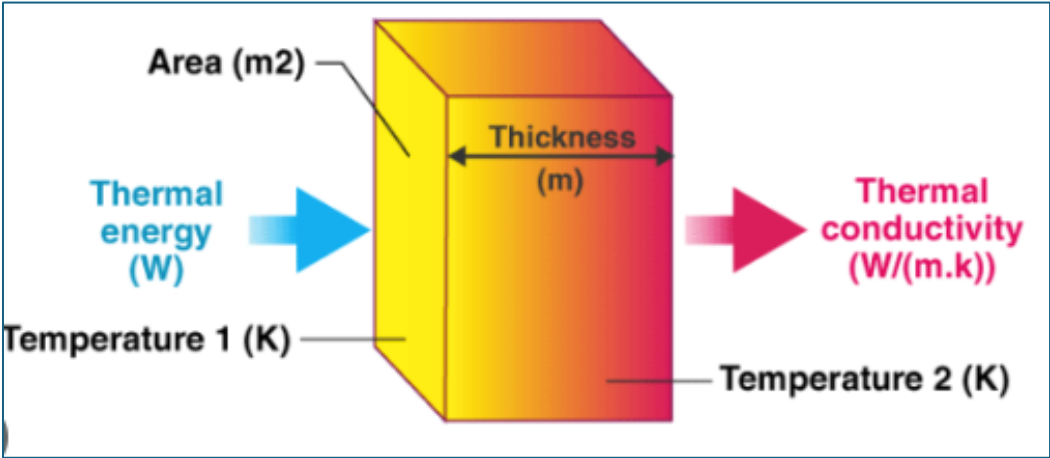


Figure 1. Thermal conductivity

Silicon as a Packaging Substrate—Applications Examples

- **2.5D and 3D Integration**
 - **Interposers**—In 2.5D integration, silicon interposers are used as a substrate to interconnect multiple dies on a single package. The interposer allows for high-density routing and integration of heterogeneous components, such as logic, memory, and analog devices.
 - **3D Stacking**—Silicon substrates with through silicon vias (TSVs) enable 3D stacking of dies, reducing the overall footprint and improving performance by shortening interconnects between layers.
- **High-Performance Computing (HPC) and AI**
 - **Data Centers and AI**—Silicon substrates are used in packaging high-performance processors and accelerators for data centers and AI applications, where thermal management, signal integrity, and high-density interconnects are critical.
 - **Graphics Processing Units (GPUs)**—Silicon substrates are also used in the packaging of GPUs, where high data throughput and efficient heat dissipation are essential.
- **RF and Microwave Applications**
 - **High-Frequency Circuits**—The low-loss characteristics of silicon substrates make them suitable for RF and microwave applications, where maintaining signal integrity at high frequencies is crucial.
 - **Antenna-in-Package (AiP)**—Silicon substrates can be used in antenna-in-package designs, integrating RF circuits and antennas on the same substrate, reducing size and improving performance.
- **Optoelectronics**
 - **Photonic Integration**—Silicon substrates can be used to integrate photonic devices with electronic circuits, enabling advanced optoelectronic packaging solutions for high-speed data communication and sensing applications.

Silicon Packaging Substrate Challenges and Considerations

- **Cost**
 - **Higher Cost**—Silicon substrates are generally more expensive than traditional organic substrates, which can be a limiting factor for cost-sensitive applications. However, the performance benefits can justify the higher cost in high-end applications.
- **Complex Manufacturing**
 - **Precision Required**—The manufacturing of silicon substrates, especially those with fine-line routing and TSVs, requires precision and advanced fabrication techniques. This can complicate the production process and increase costs.
- **Fragility**
 - **Brittleness**—Silicon is brittle and can be prone to cracking under mechanical stress, particularly during the handling and assembly processes. Careful design and process control are needed to mitigate this risk.

Take Away

Silicon as a packaging substrate offers several advantages, particularly for high-performance and high-density applications. Its excellent thermal conductivity, electrical performance, and compatibility with advanced packaging techniques like TSVs and 3D integration make it an ideal choice for cutting-edge semiconductor devices. Despite challenges such as cost and fragility, the benefits of using silicon substrates in applications like high-performance computing, RF, and optoelectronics often outweigh these drawbacks, making it a key material in the evolution of semiconductor packaging.